

Addressing Modes Examples

Register Addressing

Instruction	Instruction: Add R4, R3
Given	R4 = 200, R3 = 500
Calculation	Operation: $R4 = R4 + R3$
Final Result	Result: R4 = 700

Immediate Addressing

Instruction	Instruction: Add R4, #3
Given	R4 = 200
Calculation	Operation: $R4 = 200 + 3$
Final Result	Result: R4 = 203

Displacement Addressing

Instruction	Instruction: Add R4, 100(R1)
Given	R1 = 1000, Mem[1100] = 40
Calculation	Effective Address = $100 + 1000 = 1100$
Final Result	Result: $R4 = 200 + 40 = 240$

Register Indirect

Instruction	Instruction: Add R4, (R1)
Given	R1 = 1000, Mem[1000] = 50
Calculation	Effective Address = 1000
Final Result	Result: $R4 = 200 + 50 = 250$

Indexed Addressing

Instruction	Instruction: Add R3, (R1 + R2)
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Given	$R1 = 1000, R2 = 20, \text{Mem}[1020] = 80$
Calculation	Effective Address = 1020
Final Result	Result: $R3 = 500 + 80 = 580$

Direct Addressing

Instruction	Instruction: Add R1, (1001)
Given	$\text{Mem}[1001] = 60$
Calculation	Operation: $R1 = 1000 + 60$
Final Result	Result: $R1 = 1060$

Memory Indirect

Instruction	Instruction: Add R1, @(R3)
Given	$R3 = 500, \text{Mem}[500] = 3000, \text{Mem}[3000] = 25$
Calculation	Effective Address = $\text{Mem}[500] = 3000$
Final Result	Result: $R1 = 1000 + 25 = 1025$

Autoincrement

Instruction	Instruction: Add R1, (R2)+
Given	$R2 = 1000, \text{Mem}[1000] = 50, \text{Word size} = 4 \text{ bytes}$
Calculation	Operation: $R1 = 1000 + 50, R2 = 1004$
Final Result	Result: $R1 = 1050, R2 = 1004$

Autodecrement

Instruction	Instruction: Add R1, -(R2)
Given	$R2 = 1004 \rightarrow 1000, \text{Mem}[1000] = 50$
Calculation	Operation: $R1 = 1000 + 50$
Final Result	Result: $R1 = 1050, R2 = 1000$

Scaled Addressing

Instruction	Instruction: Add R1, 100(R2)[R3]
Given	$R2 = 1000, R3 = 5, \text{element size} = 4 \text{ bytes}, \text{Mem}[1120] = 30$

Calculation	Effective Address = $100 + 1000 + (5 \times 4) = 1120$
Final Result	Result: $R1 = 1000 + 30 = 1030$